

PROJECT CASE STUDY

KontrolAlt

A production creator-discovery and growth-intelligence platform built to find, score, and monitor emerging channels across alternative media platforms — at scale, without manual searching.

SYSTEM

**Full-Stack Web
Platform**

SCOPE

**Frontend • Backend •
Scraper • Infra**

STATUS

**Deployed & Operating in
Production**

01 The Problem

Businesses that run creator partnerships, sponsorships, or affiliate programs face the same bottleneck: finding the right channels to work with is a slow, manual, and inconsistent process.

Scouting is typically done by hand — scrolling platforms, guessing which channels are actually growing, and losing track of who's already been contacted. It doesn't scale past a handful of team members, it produces no consistent record of engagement quality, and it's blind to platforms that resist automated access in the first place.

KontrolAlt was built to replace that manual process with a system that continuously discovers channels, scores them on real engagement data, flags the ones worth pursuing, and keeps a running record of outreach status — so decisions are made on data, not guesswork.

02 What Was Built

KontrolAlt is a complete, four-part system — not a script or a prototype. Each part was designed, built, and deployed independently, then wired together into one production platform.

DISCOVERY & SCRAPING ENGINE

Automatically finds and scrapes creator channels across Rumble and Substack, extracting subscribers, engagement, posting cadence, and content data — even from platforms actively designed to block bots.

ANALYTICS & INTELLIGENCE LAYER

Turns raw scraped data into decisions: growth-velocity trends, AI-generated channel summaries, engagement-tier classification, and a "similar channels" finder to widen a prospect list from a single seed channel.

CLIENT-FACING DASHBOARD

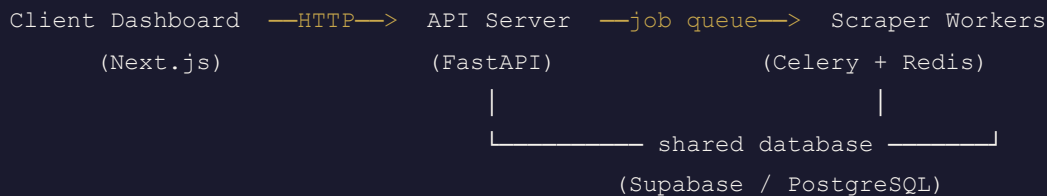
A searchable, filterable, sortable interface for browsing every discovered channel, exporting shortlists to CSV/Excel, and drilling into a full profile for any individual creator.

OPERATIONS CONTROL PANEL

An admin console for triggering scrape jobs on demand, monitoring the health of every background worker in real time, and managing the business rules (competitor lists, category taxonomies) that drive classification.

03 How It Works

The system is built as four coordinated services, each with a single responsibility. This separation is a deliberate design choice — it means the part that automates web browsers (the highest-risk, most failure-prone piece) never touches the API or the database directly, and can be scaled, retried, or replaced without putting the rest of the system at risk.



Discovery pipeline

New channels enter the system three ways: automated keyword-and-seed search (powered by a search API), bulk URL import for large batches, and a fuzzy name resolver that finds the right channel from just a creator's name — useful when the exact URL isn't known.

Resilient scraping

Alternative media platforms actively fight automated access. The scraper runs real, stealth-configured browser sessions with rotating residential proxies pinned per channel, and includes dedicated logic to detect and retry through Cloudflare's bot-verification challenge — so scrape jobs recover automatically instead of silently failing.

Scheduled & on-demand orchestration

Scrape jobs run as multi-step background workflows (chains of dependent tasks) dispatched with staggered timing to avoid overloading any single platform or proxy pool, coordinated through a message queue so the API server is never blocked waiting on a scrape to finish.

Data & intelligence

Every scraped snapshot feeds a shared database used by both the scraper and the API. On top of that raw data sits an intelligence layer: growth-velocity scoring (30- and 90-day trend comparisons), AI-generated narrative summaries of each channel's content and audience, and a lookalike-matching algorithm that surfaces similar channels from topic tags and engagement profile.

04 Key Features

- ✓ **Multi-platform discovery** — Rumble and Substack today, with an architecture that extends cleanly to additional platforms.
- ✓ **Advanced filtering & search** — filter by platform, engagement tier, topic category, subscriber range, view/comment ranges, and last-active date, all combinable at once.
- ✓ **One-click export** — any filtered view can be exported straight to CSV or Excel for outreach and reporting.
- ✓ **AI channel intelligence reports** — auto-generated narrative summaries covering content focus, audience profile, and partnership fit for every channel.
- ✓ **Growth velocity tracking** — 30-day and 90-day view/comment trend scoring shows which channels are accelerating, not just which are currently large.
- ✓ **Lookalike / similar-channel finder** — expand a shortlist from any single channel using topic and engagement-based matching.
- ✓ **Outreach status tracking** — per-channel partnership state (current partner, previously hired, do-not-contact) keeps outreach history consistent across a team.
- ✓ **Live operations console** — real-time worker health monitoring, manual job triggers, and a full audit log of every administrative action.
- ✓ **Secure, role-based access** — authenticated login with a separate admin tier for operational controls.

05 Technical Approach

For technical readers: the system was built with production reliability and clean separation of concerns as first-order priorities, not afterthoughts.

LAYER	STACK	PURPOSE
Frontend	Next.js 16 (App Router), TypeScript, Tailwind CSS	Dashboard, channel detail, admin console
API	FastAPI (Python), Supabase Auth (JWT)	Business logic, authentication, cross-service dispatch
Background workers	Celery, Redis	Scrape orchestration, scheduled & on-demand jobs
Scraping engine	Playwright (Python), residential proxy pool	Stealth browser automation, Cloudflare-bypass, retry logic
Data	Supabase (PostgreSQL)	Shared source of truth for both API and scraper
Infrastructure	Vercel, Render (Docker), Upstash	Frontend hosting, containerized services, managed Redis

Design decisions that matter

- **Strict service boundaries.** The API never imports scraper code, and the frontend never talks to the scraper directly — every cross-service interaction goes through a typed HTTP contract or a job queue. This keeps a failure in one service (e.g., a platform changing its page structure) from cascading into the others.
- **Anti-bot resilience by design.** Proxy health scoring and automatic quarantine, per-channel proxy pinning for session consistency, and dedicated Cloudflare-challenge detection with a two-cycle retry — built specifically for platforms that were not designed to be scraped.
- **Operational visibility.** Every scrape attempt is logged with status and error detail; every admin action is recorded in an audit trail; every background worker reports live health to the admin console.
- **Built to extend.** Adding a new platform means adding a new scraper task module and a new worker queue — the discovery, dashboard, analytics, and admin layers require no changes.

06 Value Delivered

This is the difference between a manual process and a system:

Continuous

DISCOVERY, NOT ONE-OFF
SEARCHES

Data-Driven

SCORING, NOT GUESSWORK

Team-Wide

SHARED OUTREACH RECORD

- ✓ **Time saved.** Channels are found, scraped, and scored automatically instead of requiring a person to search platform-by-platform.
- ✓ **Better decisions.** Growth velocity and engagement-tier data surface channels that are actually gaining traction — not just the largest ones already saturated by other sponsors.
- ✓ **No duplicated outreach.** A shared partnership-status record (current partner, previously hired, do-not-contact) prevents overlapping or repeated outreach across a team.
- ✓ **Faster shortlist expansion.** The lookalike finder turns one good channel into a list of comparable prospects in a single click.
- ✓ **Confidence in the data.** Scrape history and health monitoring make data gaps visible and explainable, instead of silent and unexplained.

KontrolAlt — a full-stack platform combining web scraping, background job orchestration, and applied data analytics into a single production system for creator discovery and growth intelligence.